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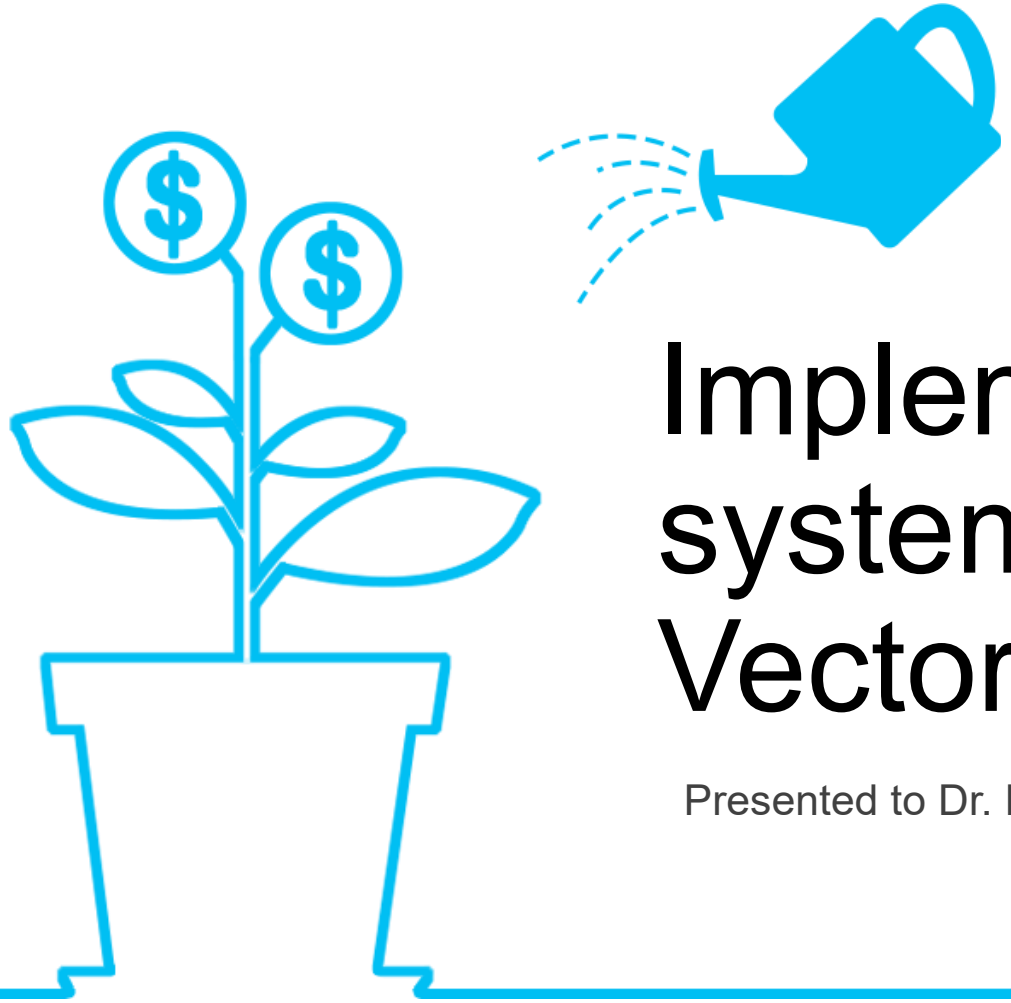


Implementation irrigation system using Support Vector Machine

Presented to Dr. Heba Draz, ECIP Internship Program Committee

Presented by: Hazem & Ali

https://eri.sci.eg

Agenda

01 Problem Statement

A Problem Statement is a clear, concise description of an issue that needs to be solved. It explains what the problem is, who it affects, and why it matters, without giving the solution yet.

02 About the Authors

A short section that gives brief background information about the people who wrote the paper.

03 Proposed Solution

The Proposed Solution explains how to solve the problem described in the problem statement.

04 Support Vector Machine

SVM is a machine learning method that learns from examples, finds patterns and classifies new data into groups.

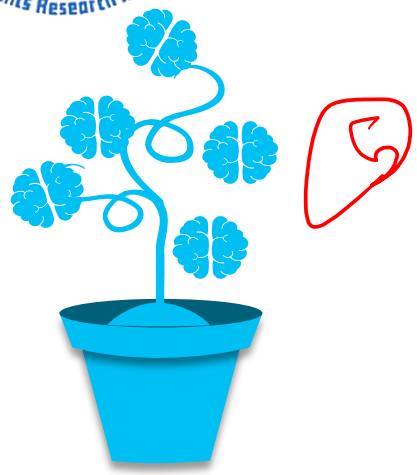
05 Communication Protocol

What is the best for real-time IoT communication?

Problem Statement

Why is it important to fix?

Problem Statement – Inefficient Irrigation in Farming



01

Farmers usually water plants manually or with simple timers

02

This causes two big problems:
Too much water → waste of water
Too little water → plants do not grow well

03

Because of climate change farmers not using technology

04

As engineers we need to solve this problem

05

The authors want a system that:

- decides how much water plants need
- opens the water pump valve by the correct amount.

About the Authors

Paper Title:

Implementation of an Irrigation System Using Support Vector Machine for Precision Agriculture Based on IoT

Authors: A Sumarudin, E Ismantohadi, A Puspaningrum, S Maulana and M Nadi

Affiliations: Informatics Department, Politeknik Negeri Indramayu, Jl. Lohbener Lama no 08, Indonesia

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DOI:10.1088/1757-899X/1098/3/032098

Proposed Solution - How does the system work?

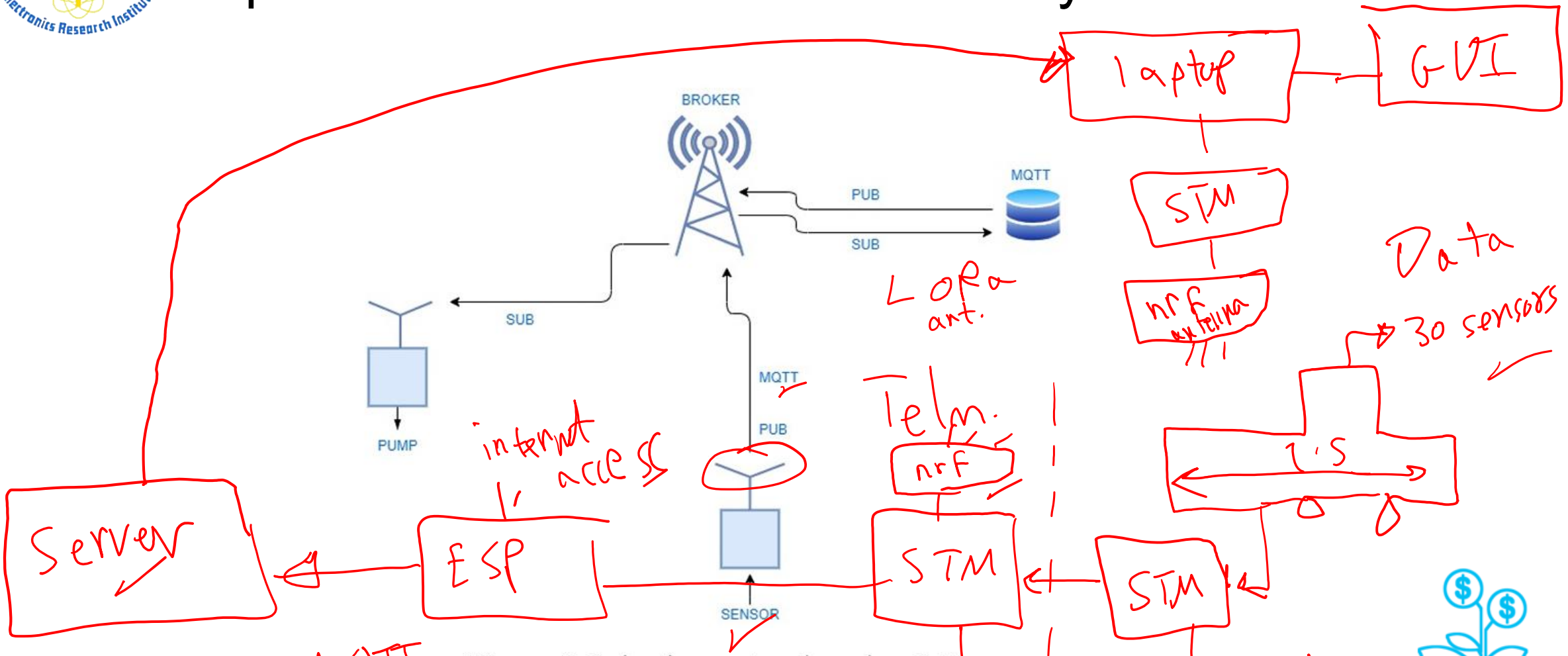


Figure 1. Irrigation system based on IoT.

Proposed Solution - How does the system work?

Actuator node opens pump valve

So watering becomes automatic and smart.

Send Data

Prediction result is sent back using MQTT.

Support Vector Machine

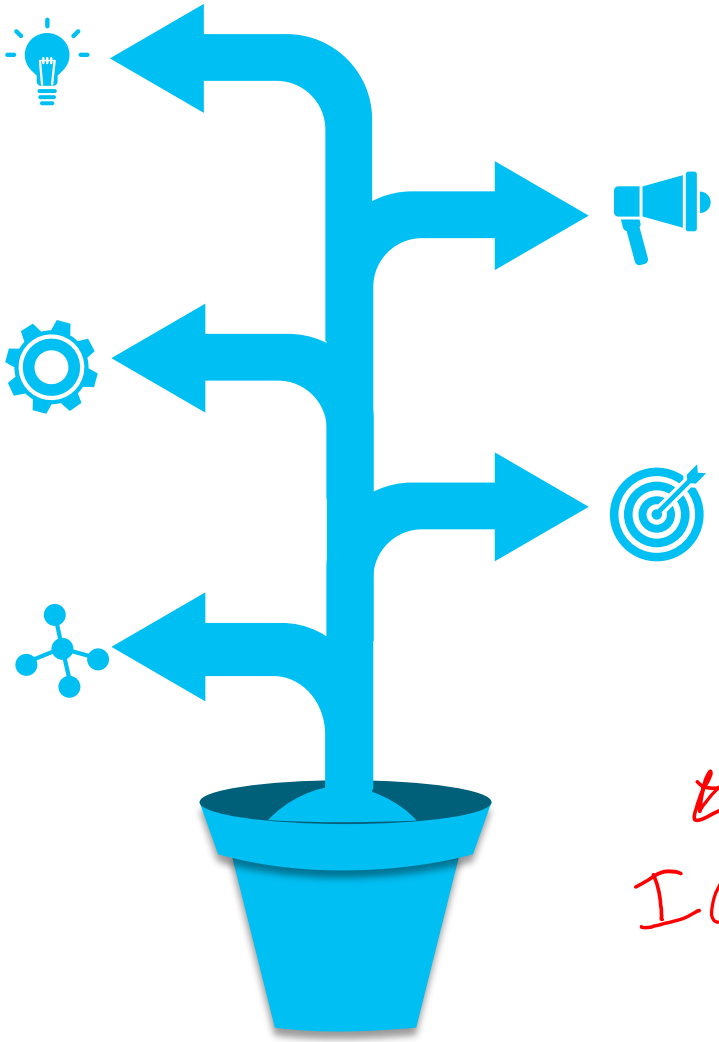
Server uses Support Vector Machine to predict how much water is needed.

Collecting Data

1. Sensor nodes collect data.
2. Data is sent wirelessly using LoRa radio.
3. Data arrives at a gateway.
4. Gateway sends data to the server using MQTT protocol.

Sensors are placed near plants.

They measure:
Soil Moisture (how wet the soil is)
Air Temperature
Air Humidity



rf antenna

IOT

FTP

What is Support Vector Machine (SVM)?

Support Vector Machine

SVM is a machine learning method that learns from examples, finds patterns and classifies new data into groups.

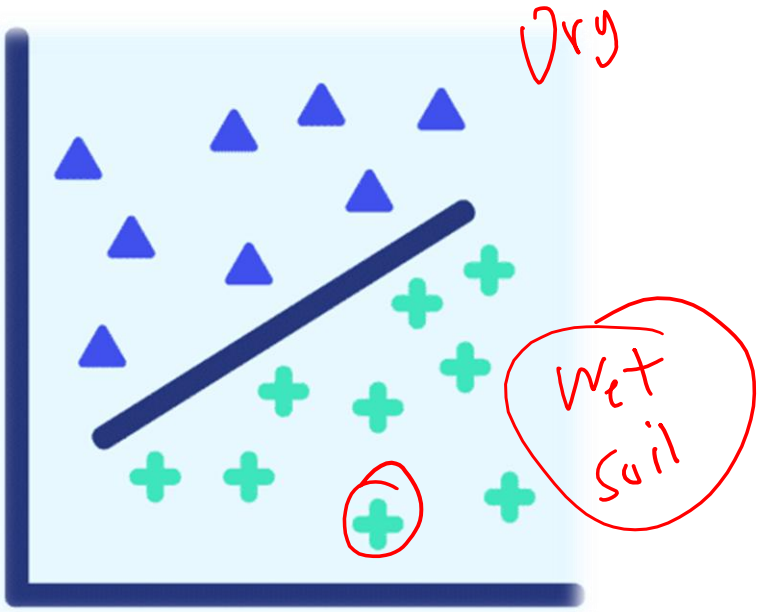
Another Definition: Support Vector Machine is a method that finds the best boundary to separate data into different groups with the maximum margin between them.

How it works?

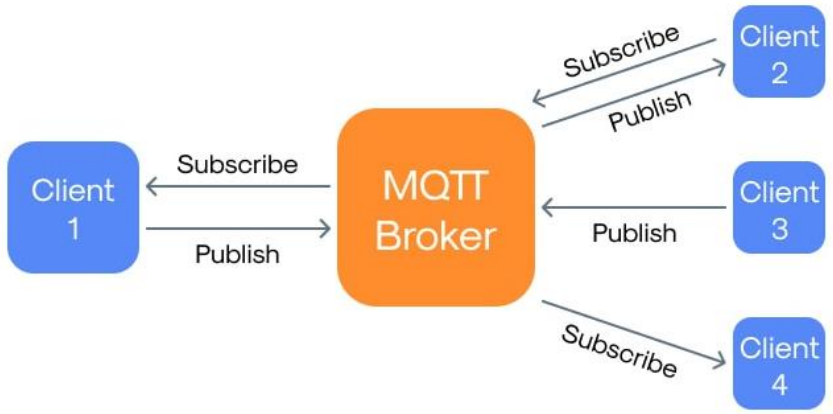
Imagine you have two types of soil data:

- Dry soil data
- Wet soil data

SVM draws a line (or curve) that separates them as clearly as possible, keeping the largest possible distance from both groups.
That line is called the hyperplane.
The closest points to that line are called support vectors and they decide the position of the boundary.

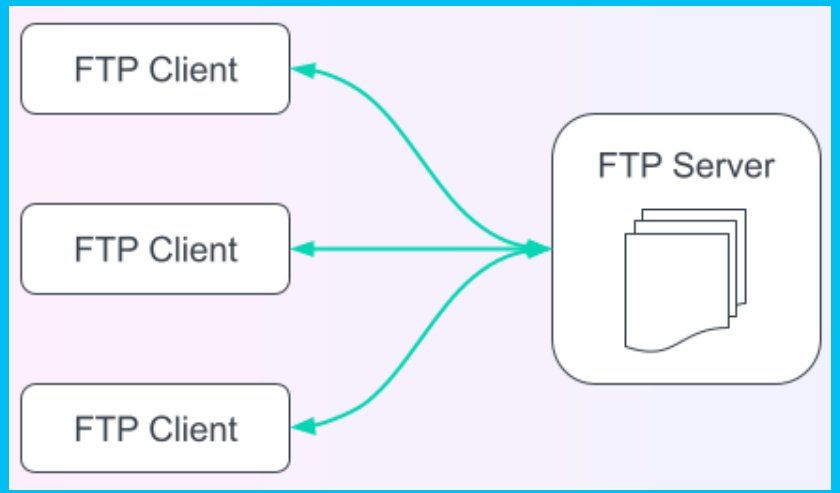


Communication Protocols



MQTT

VS



FTP

Communication Protocol

01 MQTT Communication Protocol

Message Queuing Telemetry Transport

Key Features:
- Publish/Subscribe model
- Low power and low data usage
- Fast and reliable message delivery
- Ideal for sensors and embedded devices

How does MQTT work?
- Publisher: Sensor nodes send data (soil moisture, temperature)
- Broker: Central server that manages messages
- Subscriber: Actuators or applications receive data and act on it

02 FTP Communication Protocol

File Transfer Protocol

Key Features
Client–server based communication
Used for uploading and downloading files
Reliable file transfer using TCP
Requires stable network connection
Not optimized for low-power or real-time systems
How Does FTP Work?
Client: Requests to upload or download files
Server: Stores and manages files
Limitations for IoT:
High bandwidth and power consumption
No real-time data streaming

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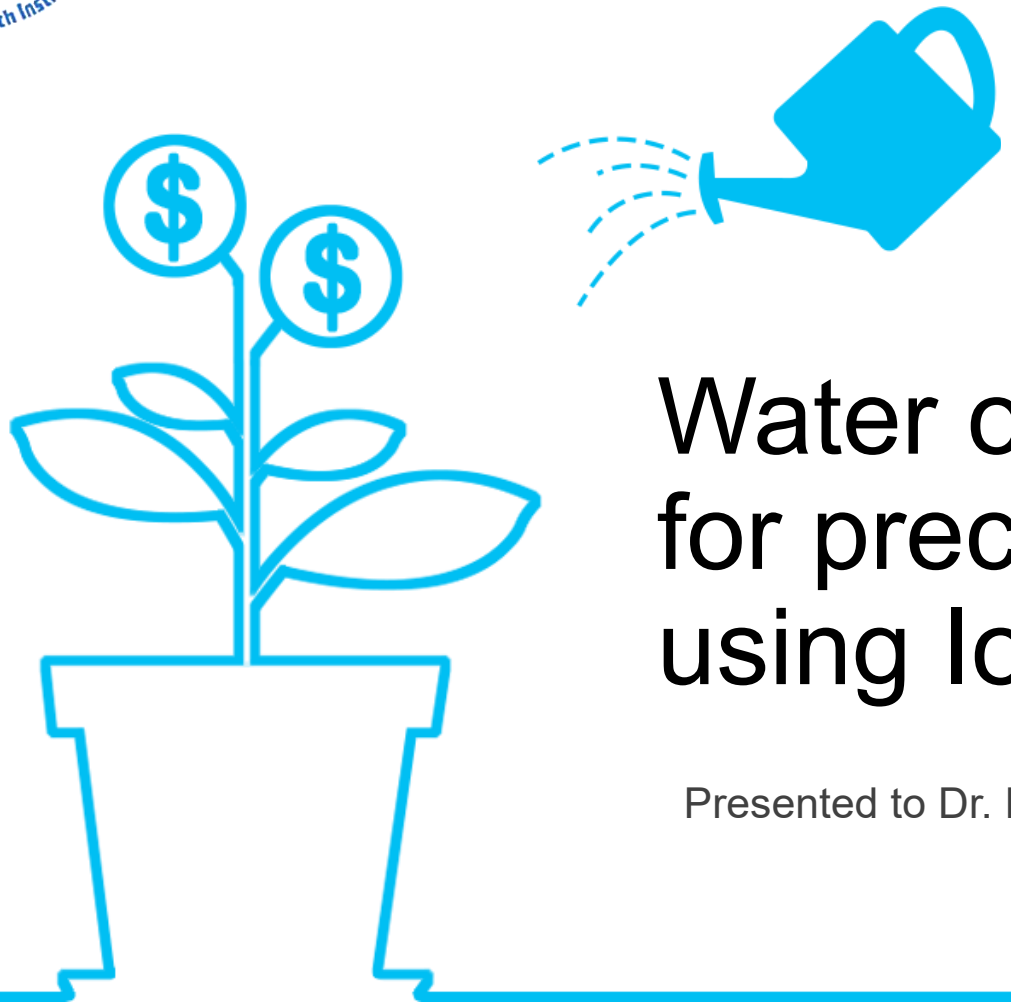

Water optimization technique for precision irrigation system using IoT and machine learning

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03 Why Banana Crop?

Banana requires large, continuous water supply

04 Overall System Architecture

The system has **3** layers

05 Role of Each Algorithm & Conclusion

K-Nearest Neighbors (KNN), Gradient Boosted Tree (GBT) and LSTM (Long Short-Term Memory)

Problem Statement

Why is it important to fix?

Problem Statement

01 Core Problem

Agriculture consumes ~70% of global freshwater and huge electricity

Traditional irrigation:

- Over-waters crops
- Wastes energy (pumps, motors)
- Is often manual or rule-based

02 Goal of the Paper

Optimize water usage and reduce energy consumption in irrigation using IoT + Machine Learning

Specifically:

- Use real-time sensor data
- Apply machine learning forecasting
- Make data-driven irrigation decisions
- Test the system on banana cultivation

About the Authors

Paper Title:

Water optimization technique for precision irrigation system using IoT and machine learning.

Affiliations:
Department of Computer Science and Engineering, Jain University, Bangalore, India
School of Computer Science, Galgotias University, Greater Noida, India
School of Information Technology and Engineering, Vellore Institute of Technology, Vellore, India

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DOI: 10.1016/j.seta.2022.102307

Why Banana Crop?

Why Banana Crop?

Banana requires large, continuous water supply
Sensitive to:

- Soil moisture
- Root depth
- Rainfall
- Humidity

Any optimization here = significant water & energy savings

Key result:

- 31.4% water saved in 2020
- ~50% water saved in 2021 (due to rainfall-aware prediction)

Overall System Architecture

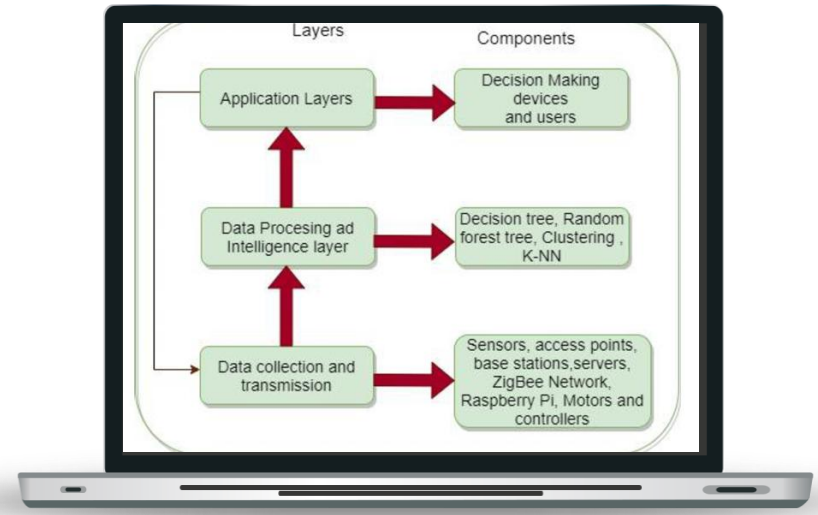
Overall System Architecture

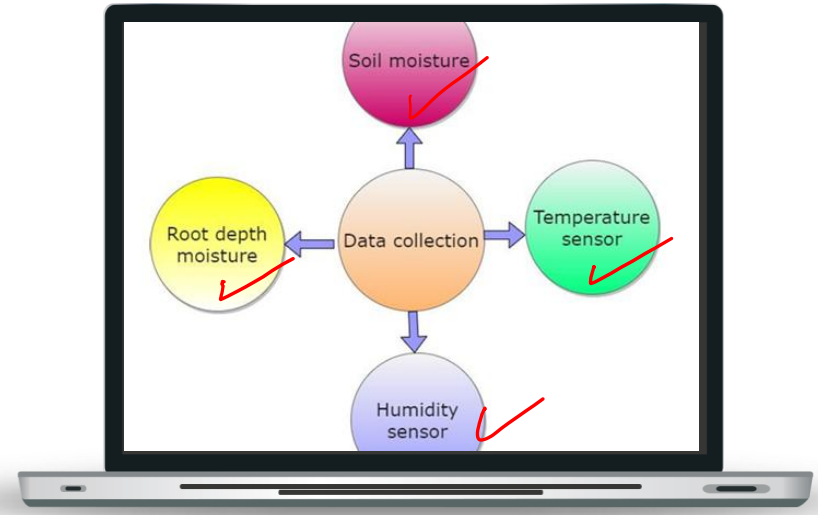
The system has 3 layers:

i. Data gathering/collection and transmission
ii. Data processing and intelligence layer
iii. Application Layers

Why a Hybrid Machine Learning Model?

- Problem with Existing Systems
  - Most use only one ML algorithm
- Fail to capture:
  - Non-linear relationships
  - Time-series patterns
- Solution in This Paper
  - Combine strengths of multiple models and select the best prediction dynamically





Technologies used

Wireless Sensor Networks (WSN)
ZigBee
Cloud storage

Sensors collect

Soil moisture
Root depth moisture
Temperature

Role of Each Algorithm

Role of Each Algorithm

RNN

Machine Leaning

K-Nearest Neighbors (KNN) 1

Purpose:
Finds nearest sensor readings
Uses data from nearby plants / zones
Why KNN?
No assumptions about data distribution
Good for local similarity
Example:
If nearby banana plants need water, this plant likely needs it too

LSTM (Long Short-Term Memory) 3

Purpose:
Time-series forecasting
Predicts future water needs
Why LSTM?
Remembers past conditions
Example:
If moisture has been dropping for 3 days, LSTM predicts higher irrigation need

Gradient Boosted Tree (GBT) 2

Purpose:
Predicts real water requirement
Handles non-linear relationships
Why GBT?
High accuracy
Strong regression performance
Combines weak decision trees into a strong model
Used for:
Moisture ✓
Temperature ✓

Spearman's Rank Correlation (SRC)

1 2 3

Purpose:
Compares predicted values vs actual values
Ranks KNN, GBT, and LSTM predictions
Why Spearman?
Chooses the best model at each time interval
Final Output:
A consolidated hybrid prediction

R^2

Conclusion

STRENGTHS

- Real-world deployment
- Multi-sensor data
- Hybrid ML (not single algorithm)
- Time-series forecasting
- Significant water & energy savings

WEAKNESS

- Not fully automated irrigation
- Tested only on banana crop
- Noise in sensor signals can affect accuracy

Future Work

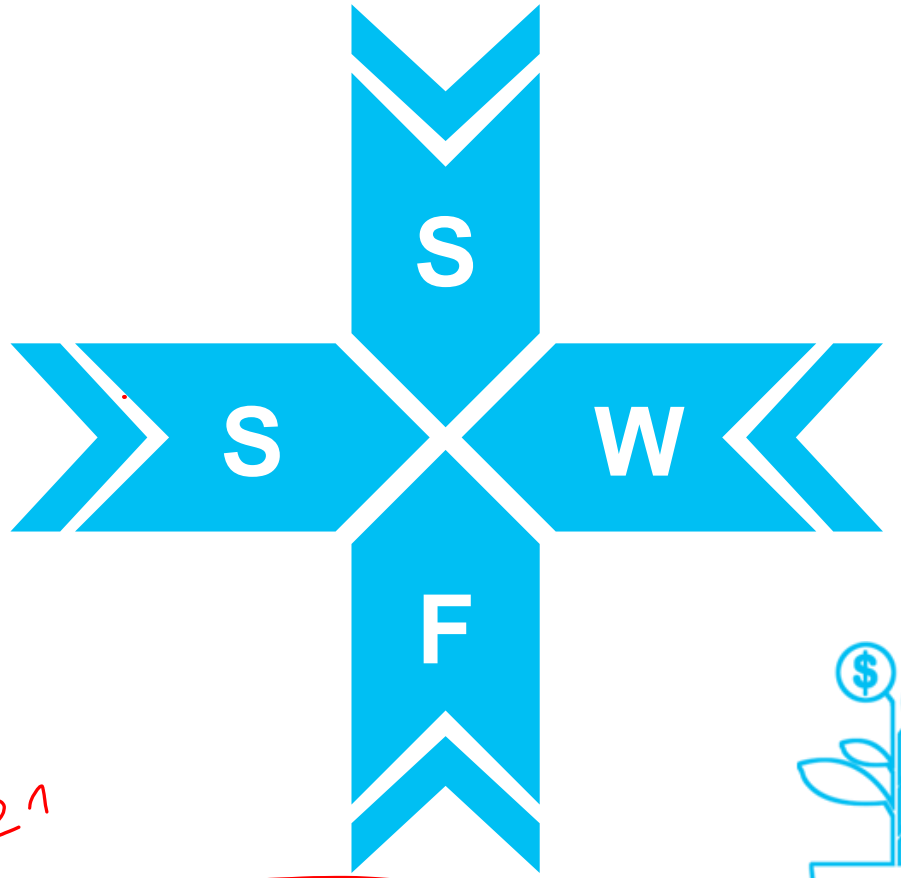
- Full automation of irrigation
- Noise-resilient models
- Apply to Carrot

Summary

This paper proposes an IoT-based hybrid machine learning system that combines KNN, GBT, LSTM, and Spearman correlation to optimize irrigation water usage, achieving up to 50% water savings in banana cultivation while indirectly reducing energy consumption.

2021

3 years



Thank you

Any Questions?

Green house ...
Irr.

Data set

python → ML
SVM